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Innovative Invincible Trim Choke Application in Unconventional Gas Wells

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Abstract

This study presents a field-proven solution to one of the most persistent operational challenges in gas production: choke valve trim erosion in unconventional gas wells. A six-month trial was performed to evaluate the application of an Invincible Solid Diamond Trim choke valve as an alternative to the conventional tungsten carbide trim choke configuration. The trial, executed on two erosion-prone wells, demonstrated that the Invincible Solid Diamond Trim design offers a significantly extended service life – at least six times longer – with an optimistic scenario of potential elimination of replacement needs over the well's lifecycle, while delivering steady gas flow, consistent pressure control, and dramatically reduced operational disruptions. The field results were substantiated by inspection, test data, and economic modeling, all of which confirmed that the innovative trim design not only improves performance but also contributes to substantial cost avoidance and enhanced safety. The findings support broader application of Invincible Solid Diamond Trim in erosive environment and encourage proactive consideration during field development planning.

Introduction

Choke valves serve as a critical control point in oil and gas wellhead assemblies, regulating the flow rate of fluids and preventing over-pressurization of downstream facilities. In unconventional reservoirs, where produced gas may carry formation fines and high-pressure gradients are common, choke trim erosion is not only frequent but also economically disruptive. Repeated choke valve failures result in increased maintenance frequency, gas production downtime, and safety hazards due to the need for high-risk interventions. Historically, tungsten carbide trim chokes have been used in erosive services; however, their service life has proven insufficient in certain highly erosive applications, such as those with high sand content and high differential pressure environments.

This paper documents the successful introduction of Invincible Solid Diamond Trim choke technology on two erosive gas production wells. The Invincible Solid Diamond Trim was primarily developed for frac plug drill-outs and flowback on unconventional wells, where it has demonstrated vastly superior trim life compared to conventional tungsten carbide trims. The technology's ability to maintain dimensional integrity

and erosion resistance, even under severe flow conditions, positions it as a transformative solution for choke erosion management in unconventional gas fields, or indeed any highly erosive throttling application.

Statement of Theory and Definitions

Choke trims in oil and gas production applications have long used solid tungsten carbide for erosive service. At its introduction, it demonstrated up to 100× greater erosion resistance compared to hardened steel trims and remains the industry standard for abrasive environments [5, 6]. The Invincible Solid Diamond Trim choke trim is constructed from solid Polycrystalline Diamond (PCD), known for its superior hardness, exceptional wear resistance, and chemical stability. PCD is the most common material used in oil and gas well drill-bits where it replaced tungsten carbide bits. PCD bits drill faster, more efficiently, and have a much longer lifespan than tungsten carbide bits. Solid PCD will typically outlast solid tungsten carbide by 50 or even 100 times in abrasive or erosive service. [3, 4]

Originally developed for use in exceptionally erosive choke applications such as frac plug drill-out and flowback operations, the Invincible Solid Diamond Trim was repurposed for continuous gas production control in this trial.

The Invincible Solid Diamond Trim retrofit was performed on existing production chokes (see Figure 1), converting them from tungsten carbide trims to solid diamond trims without modification of the valve body or actuator (see Figure 2).

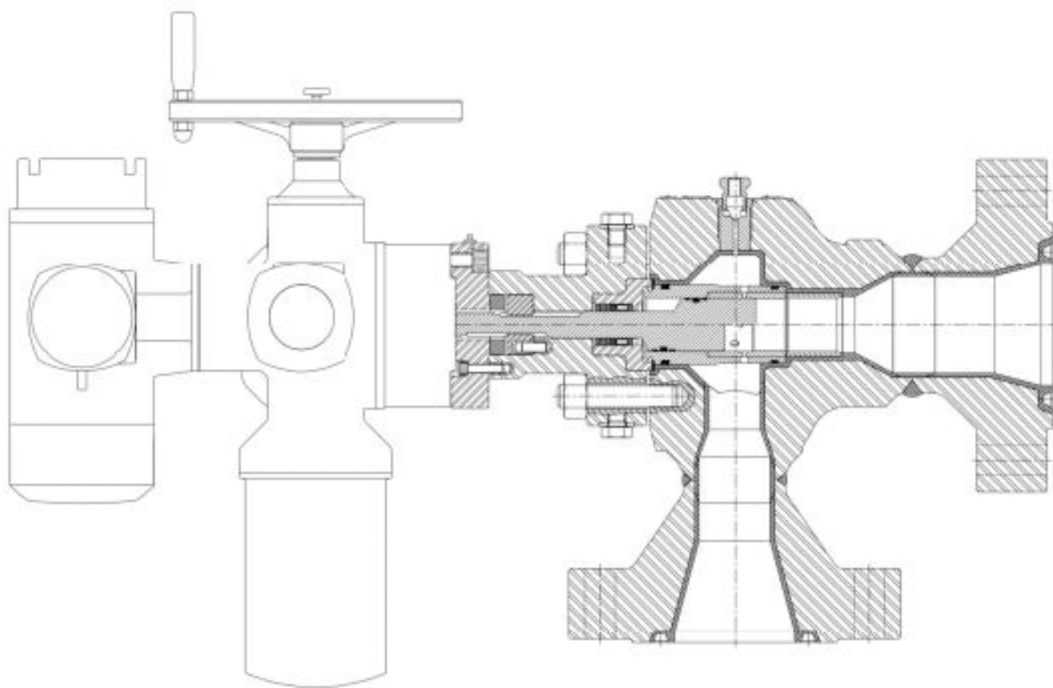


Figure 1—Existing production choke with tungsten carbide trim

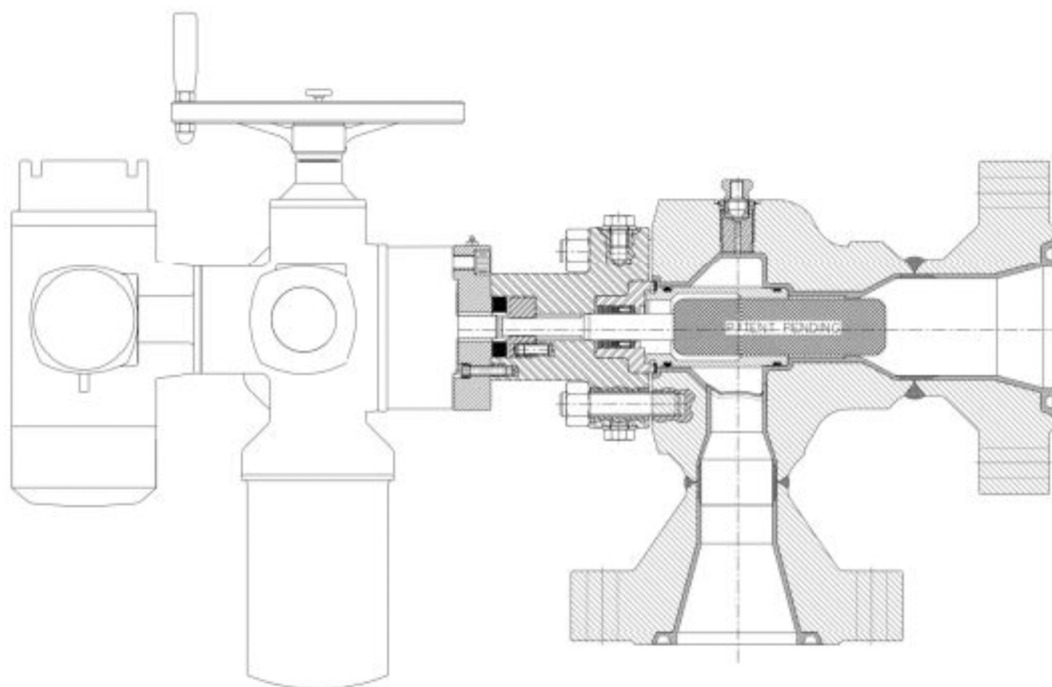


Figure 2—Existing production choke retrofitted with invincible solid diamond trim

Description and Application of Equipment and Processes

This comprehensive field trial was strategically executed, meticulously targeting two of the most erosion-prone wells within our operational portfolio: Well-A and Well-B. These specific wells were deliberately selected due to their well-documented history of severe and rapid choke trim degradation. Prior to the commencement of this trial, both Well-A and Well-B were notorious for exhibiting an exceptionally short lifespan for conventional choke valve trims, consistently necessitating costly and disruptive trim replacements every two to four months. This frequent maintenance cycle not only imposed significant financial burdens through material and labor costs but also resulted in considerable production downtime, directly impacting overall operational efficiency and revenue generation. The pervasive issue of premature trim failure in these challenging environments underscored an urgent need for a more durable and resilient solution.

The core of this trial involved the retrofitting of existing choke valve bodies with the innovative Invincible Solid Diamond Trim technology. This advanced trim was installed into the choke valves of both Well-A and Well-B, replacing the less robust traditional materials. Following installation, these newly equipped chokes were subjected to an intensive monitoring period spanning six consecutive months. This extended monitoring duration was designed to provide a statistically significant dataset, allowing for a thorough evaluation of the new trim's long-term performance under actual field conditions, encompassing various flow regimes and environmental stressors typical of these wells.

Throughout the six-month trial, a rigorous data logging protocol was implemented to capture a comprehensive array of operational parameters on an hourly basis. Key metrics meticulously recorded included the gas production rate, measured in Million Standard Cubic Feet per Day (MMSCFD), which served as a direct indicator of well output and stability. The choke valve opening percentage was also precisely logged, providing insight into the control valve's positioning and its responsiveness to flow demands. Furthermore, upstream and downstream pressures were continuously monitored, enabling detailed analysis of pressure differential across the trim and its impact on erosion. Flowing temperatures were additionally recorded, as temperature fluctuations can significantly influence fluid properties and,

consequently, trim wear mechanisms. The aggregation of this detailed operational data allowed for a holistic understanding of the system's behavior and the trim's performance under dynamic conditions.

The evaluation of the Invincible Solid Diamond Trim's effectiveness was primarily centered on three critical performance indicators: flow consistency, trim integrity, and control stability. Flow consistency assessed the ability of the choke to maintain a steady and predictable flow rate without erratic fluctuations, a common issue with eroded trims. Trim integrity was evaluated by closely monitoring for any signs of wear, deformation, or material loss. Control stability measured the choke valve's responsiveness and precision in maintaining desired pressure and flow set points, minimizing oscillations and operational deviations. The culmination of the trial involved a comprehensive end-of-trial teardown and detailed internal examination of the retrofitted chokes. This meticulous inspection involved dismantling the valves and thoroughly assessing the internal components for any evidence of erosion, cavitation, or other forms of mechanical wear. These post-trial analyses, combined with the extensive operational data, provided irrefutable evidence regarding the long-term durability and superior performance of the Invincible Solid Diamond Trim in highly erosive environment.

Presentation of Data and Results

The field trial demonstrated the exceptional performance and durability of the Invincible Solid Diamond Trim. Throughout the entire 180-day monitoring period, the diamond-enhanced choke valves performed with remarkable consistency, exhibiting no loss of flow control, visible signs of wear, or any need for operational adjustments or interventions. This stable performance stands in stark contrast to the frequent maintenance demands of traditional trims. Crucially, the relationship between flow capacity and valve opening remained perfectly stable for the entire six months. This stability allowed operators to maintain precise control over production rates and pressures, a critical factor for optimizing well performance as per subsurface guidance. The consistent flow control provided a significant operational advantage, eliminating the unpredictable behavior typically associated with degrading conventional trims.

Notably, under identical and equally challenging well conditions, traditional tungsten carbide trims exhibited a rapid and predictable decay curve. Their initial operational efficiency, typically observed with a choke opening of 30-35%, would progressively and rapidly deteriorate. Within a mere four months, this opening percentage would effectively render the choke trim fully eroded, leading to a state where flow control became virtually impossible. This dramatic loss of control often resulted in uncontrolled flow, potential well damage, and significant production disruptions, underscoring the severe limitations of conventional materials in erosive environment.

Result Summary

The comprehensive data analysis, supported by detailed trend plots, summarized in the accompanying table, revealed the following compelling key insights:

1. **Limited Lifespan of Conventional Trims:** Traditional tungsten carbide trims demonstrated an average service life of a mere 64 days, with even the most resilient units failing within a maximum of 120 days. This short lifespan necessitated frequent, costly, and disruptive replacements, directly hindering sustained production.
2. **Unprecedented Durability of Diamond Trims:** In stark contrast, the Invincible Solid Diamond Trims surpassed the 180-day trial period without exhibiting any visible signs of erosion, cavitation, or material degradation. This sustained performance far exceeded the capabilities of any conventional trim tested under similar conditions.
3. **Immediate Reversion to Old Problems:** As a critical point of comparison, a standard tungsten carbide choke was temporarily installed when the Invincible Solid Diamond Trim choke was removed

for a scheduled shop inspection. Within just 120 days, this tungsten carbide choke followed the exact same rapid erosion pattern observed historically, becoming fully eroded and confirming the severity of the well environment and the superior resistance of the diamond trim.

4. **Sustained Long-Term Performance:** The same Invincible Solid Diamond Trim, after undergoing a thorough inspection, was reinstalled into service. It has continued to demonstrate its remarkable resilience, showing no signs of deterioration even after the initial 180-day trial (Phase 1) combined with an additional 150+ days of operation in Phase 2. This extends its total operational life to over 330 continuous days, all while maintaining the same exceptional control and performance observed from day one. This unparalleled longevity represents a paradigm shift in choke trim technology.

Erosion Visualization

The degradation of traditional tungsten carbide trims due to erosion follows a disturbingly consistent and predictable pattern, one that significantly impacts operational efficiency and control. Initially, as high-velocity abrasive particles within the fluid stream continuously impinge upon the trim's surfaces, microscopic material removal begins. This abrasive wear is particularly pronounced in the most critical flow path areas, such as the cage's orifice and plug end surfaces, leading to an enlargement of the flow area. This enlargement directly results in a higher actual flow coefficient (C_v) than the design intent for a given choke opening. In practical terms, this means that for the same physical choke opening, a greater volume of fluid will pass through.

To counteract this increasing flow capacity and maintain the desired gas production rate, operators are compelled to continuously adjust the choke to a progressively lower opening percentage. This constant necessity for downward adjustment is a clear operational burden, demanding vigilant monitoring and frequent intervention. As the erosion advances relentlessly, the choke's ability to fine-tune the flow diminishes, becoming increasingly sluggish and less precise. This ultimately culminates in a complete loss of controllability, reaching a critical point where the choke valve can no longer effectively restrict flow or maintain the target production rate, regardless of how far it is closed. At this stage, the trim is essentially fully compromised, rendering the choke valve ineffective for its primary purpose of precise flow and pressure management, leading to significant operational challenges and the inevitable need for costly replacement.

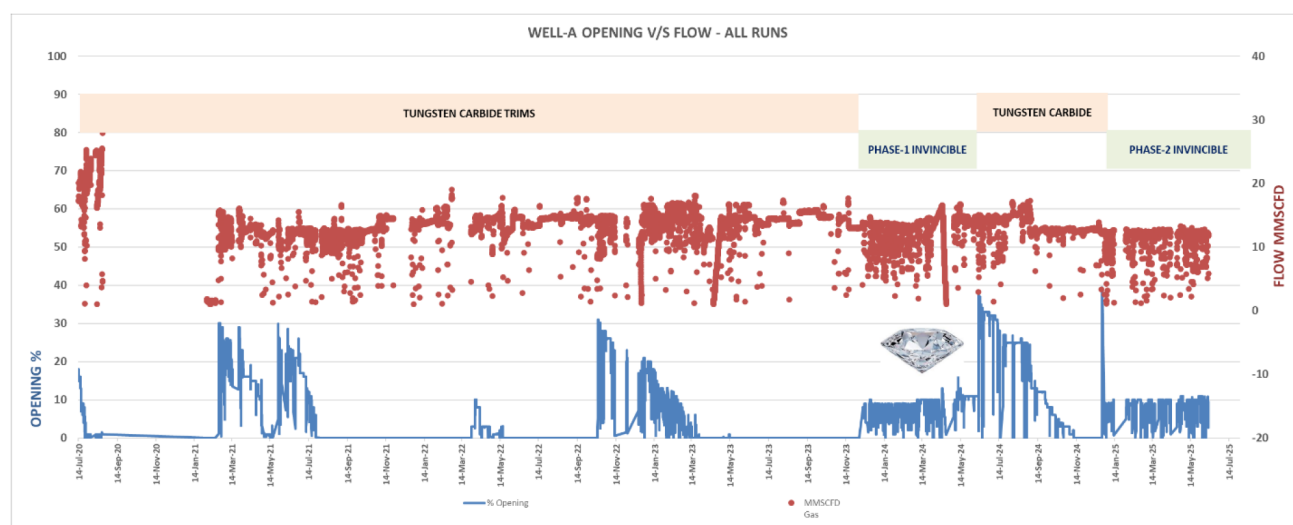


Figure 3—Opening v/s Flow for all runs

Table 1—Useful life for each trim

Run #	Trim/Runs	Days
1	(1) Tungsten Carbide, Cv-64	14
2	(2) Tungsten Carbide, Cv-13.8	69
3	(3) Tungsten Carbide, Cv-13.8	59
4	(4) Tungsten Carbide, Cv 13.8	25
5	(5) Tungsten Carbide, Cv 13.8	100
6	(Phase-1) Invincible (6 months trial), Cv=19	180
7	(6) Tungsten Carbide, Cv = 13.8	120
8	(Phase-2) Invincible (Re-installation after inspection), Cv=19	150+

Comparative Trim Performance Visualization

The accompanying chart, specifically referenced as [Figure-4](#), serves as a powerful visual representation, directly comparing the operational lifespan of choke trims before they become entirely uncontrollable. This graphical depiction clearly illustrates the pronounced difference in durability between conventional and advanced materials. For tungsten carbide trims, the data reveals a critical deterioration in choke opening over time, indicating a progressive loss of control. Their functionality consistently degrades, typically reaching a critical 0% effective opening—meaning complete loss of flow regulation, within a maximum of 120 days of service. This signifies their rapid obsolescence in challenging environments.

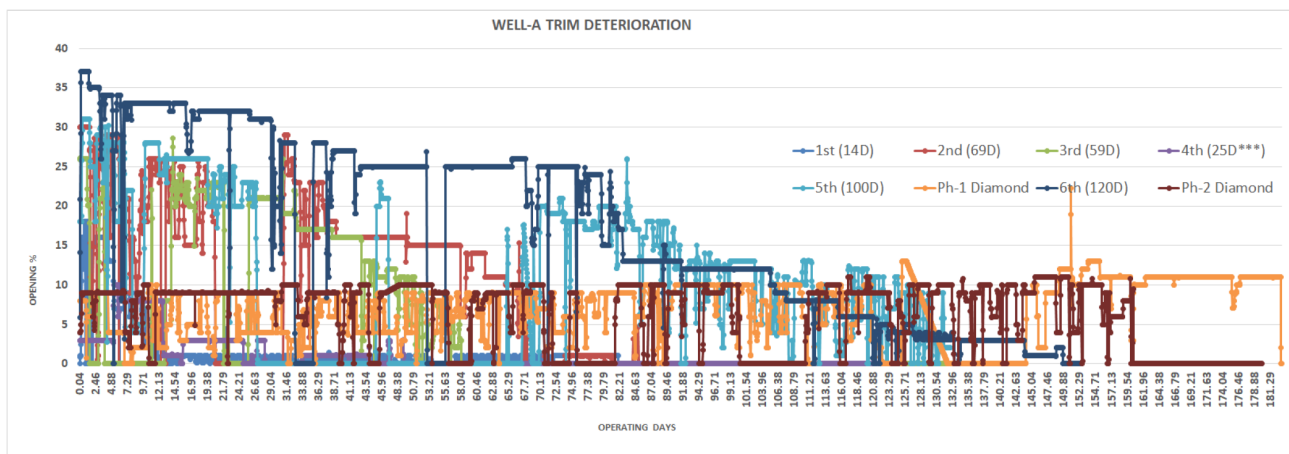


Figure 4—Trim Deterioration over days

In stark contrast, the Invincible Solid Diamond Trim exhibits a remarkably stable opening trend throughout its extensive operational period. This unwavering stability signifies no measurable deterioration whatsoever in its performance or physical integrity, not only during the initial 180-day trial but also during an additional 150 days following its re-installation post-inspection. This unprecedented consistency, spanning a total of 330 days and still counting, demonstrates the Invincible Solid Diamond trim's inherent ability to maintain precise control without succumbing to the erosive forces that cripple traditional alternatives. The comparison vividly underscores the diamond trim's superior longevity and reliability.

Monthly Deterioration and Control Challenges

The visual data presented in [Figure-5](#) powerfully depicts the severe monthly deterioration in choke opening vs. flow for traditional tungsten carbide trims, highlighting a critical loss of operational control over time. Initially, to manage the desired flow rate, the tungsten carbide trim's opening frequently fluctuates within a

15% to 35% range. This wide band of adjustment is a direct consequence of the trim continuously eroding, forcing operators to constantly compensate for the increasing flow capacity by closing the choke further. This reactive adjustment process is both imprecise and highly demanding.

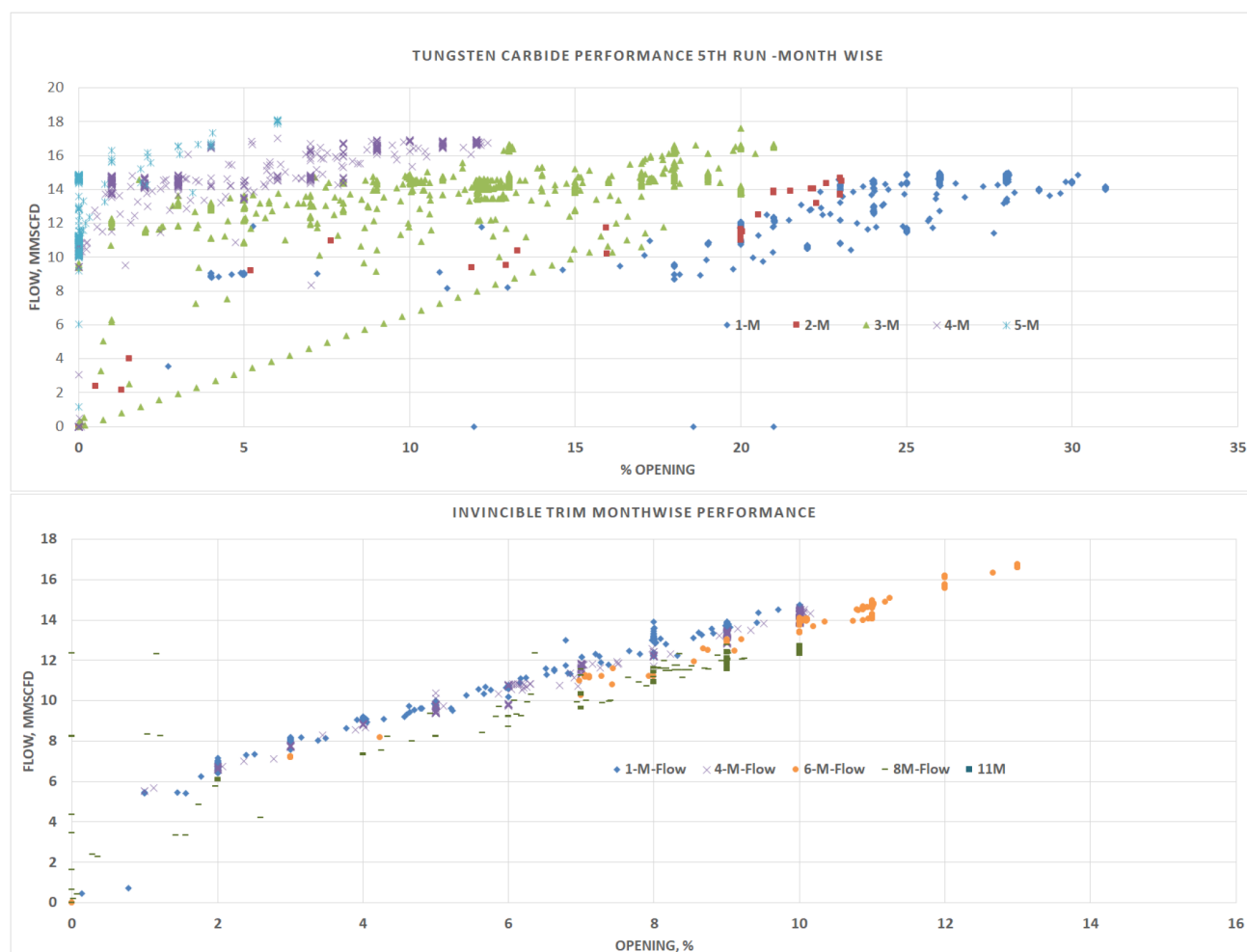


Figure 5—Deterioration pattern for tungsten carbide v/s invincible trim

As the second month of operation progresses, the effective opening range dramatically constricts to 20% to 25%, indicating significant material loss and reduced adjustability. By the third month, the situation worsens, with the controllable range narrowing further to a mere 5% to 20%. The fourth month sees the trim's functional window shrink drastically to 0% to 12%, signifying a near-total collapse of fine-tuned control. Ultimately, by the fifth month, the choke opening moves almost entirely to 0%, confirming a state of a fully eroded choke trim and a complete loss of controllability. At this point, the valve can no longer effectively restrict or regulate the fluid flow, making precise production management impossible and often necessitating immediate, unplanned shutdowns for replacement.

Graphical comparisons between choke opening and gas flow clearly illustrated this degradation, whereas Invincible Solid Diamond Trim plots remained linear and steady, allowing for consistent controllability throughout the operation. This indicates that both erosion resistance and geometrical stability are critical factors for any production choke technology.

A similar performance was observed in Well-B. Hence the charts are not repeated.

Economic Analysis

The economic implications of choke trim performance are substantial, directly impacting the profitability and operational continuity of gas production. The analysis clearly shows that traditional tungsten carbide trims installed on these challenging wells provided an average of only 64 days of service before complete loss of gas production control due to erosion. This degradation was remarkably swift, with observable erosion and the need for manual choke adjustments beginning within just a day of start-up to maintain target gas production rates. This constant intervention demands significant operator attention and is prone to human error, further exacerbating operational inefficiencies.

In stark contrast, the Invincible Solid Diamond Trim has demonstrated no visible erosion or performance degradation even after an impressive 330 days of operation. This unprecedented longevity translates into an improvement in erosion resistance conservatively estimated to be in the order of 50 to 100 times better than tungsten carbide. This level of advancement is consistent with published improvements seen in other demanding applications within the oil and gas sector, such as the widely recognized enhanced performance of Polycrystalline Diamond Compact (PDC) drill bits. The sheer durability of the diamond trim fundamentally reshapes the cost-benefit landscape for well management.

Considering the short lifespan of tungsten carbide trims, each well would necessitate at least three full trim replacements within a single year. This leads to not only significant direct costs associated with trim materials and specialized labor but also considerable indirect expenses. These indirect costs include crucial factors such as deferred production (lost revenue during downtime) and the inherent shutdown/start-up costs, which encompass personnel mobilization, safety protocols, and the energy required to bring the well back online. These cumulative expenses quickly overshadow the initial material cost of the trim itself.

Conversely, while the Invincible Solid Diamond Trim carries a higher upfront cost, its superior performance and longevity offer immense long-term value. Based on current data, it is confidently estimated to last at least two years – representing a six-fold increase over the current useful life of carbide trims. More profoundly, a potential for substantial cost savings over the entire lifecycle of multiple wells has been definitively identified. This breakthrough technology holds the remarkable potential that a single Invincible Solid Diamond Trim could effectively last for the entire operational lifespan of the well, thereby eliminating recurring replacement costs, maximizing uptime, and significantly enhancing the economic viability of complex production assets. This transforms valve maintenance from a frequent, reactive expense into a rare, predictable investment.

Health, Safety, and Environmental (HSE) Impact

The maintenance of conventional choke valves, characterized by their frequent need for replacement, inherently involves a series of complex and high-risk procedures. Each intervention typically mandates a planned shutdown of the well, complete depressurization of the line, intricate mechanical disassembly of heavy valve components, and the precise handling of lifting equipment. Every one of these steps introduces elevated risk profiles, including the potential for dropped objects, exposure to hazardous chemicals, and the inherent dangers associated with working on pressurized systems.

The necessity for frequent interventions—two to three times annually per well with traditional trims—significantly increases personnel exposure to safety hazards. This includes direct contact with high-pressure gas, the ever-present risk of potential hydrocarbon leaks during line breaks, and the pervasive threat of manual lifting injuries or more severe incidents involving heavy machinery. These repetitive tasks not only heighten the immediate risk of accidents but also contribute to long-term fatigue and stress for field personnel.

The extended lifespan offered by the Invincible Solid Diamond Trim dramatically alters this risk landscape by effectively removing two to three such high-risk interventions per well per year. This directly translates to measurable reductions in the volume of critical administrative work, such as permit-to-work applications, equipment opening and line break work permits, and hot work permits, significantly

streamlining safety compliance. Furthermore, it minimizes the need for personnel and equipment travel to remote locations, reducing road-related incidents and fuel consumption. The lessened demand for water tankers for hydrotesting or hydrotest manifolds also contributes to operational simplification and reduced logistical footprint. From a broader sustainability standpoint, fewer replacements directly lead to reduced material waste being sent to landfills, minimized vendor mobilizations (reducing associated vehicle emissions and fuel consumption), and consequently, a tangible lower carbon footprint for the entire operation. This innovative solution offers profound advantages in enhancing safety, reducing environmental impact, and promoting more sustainable operational practices.

Conclusion

The exceptional performance and reliability of the Invincible Solid Diamond Trim choke have been demonstrated in two of our most challenging wells, where it has been subjected to erosive gas production environments. Sustained erosion resistance, superior flow control, and remarkable material integrity have been achieved over trial duration of six months plus with total of 11 months continuous service.

Significant cost savings have been revealed through the analysis, resulting from reduced maintenance frequency, labor, and deferred production. Additionally, safety has been enhanced by minimizing well shutdowns and manual handling, while environmental sustainability has been promoted through reduced waste and emissions.

The incorporation of the Invincible Solid Diamond Trim into all upcoming projects and new well designs is strongly recommended, particularly for those with erosive fluid streams or fine sand presence. Its proven reliability and long-term value have made it an essential solution for optimizing production and enhancing operational safety in the most aggressive applications.

Nomenclature

PCD = Polycrystalline Diamond

C_v = Flow Coefficient

HSE = Health, Safety, and Environment

MMSCFD = Million Standard Cubic Feet per Day

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